

Applied Technology School

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The First Unit Sequences

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Unit one
Sequences

Dear student By the end of this unit you should be able to:-

- ❖ Identify the arithmetic sequence.
- ❖ Deduce the general term of an arithmetic sequence.
- ❖ Find the arithmetic mean of an arithmetic sequence.
- ❖ Find the sum of a finite number of the terms of an arithmetic sequence.
- ❖ Identify the geometric sequence.
- ❖ Deduce the general term of the geometric sequence.
- ❖ Find the geometric mean of a geometric sequence.
- ❖ Find the sum of a finite number of the terms of a geometric sequence.
- ❖ Find the sum of an infinite number of the terms of a geometric sequence.

Lesson one
Arithmetic sequence

The sequence T_n is called Arithmetic sequence If :-

$$T_{n+1} - T_n = \text{constant} \quad \forall n \in \mathbb{Z}^+$$

- In the arithmetic sequence the difference between a term and the directly previous term to it equals a constant.
- This constant is called the common difference of the sequence, and it is denoted by the symbol (d).

Example 1:-

- (5 , 9 , 13 , 17 ,) is an arithmetic sequence its common difference = 4
- (-11 , - 9 , - 7 , - 5 ,) is an arithmetic sequence its common difference = 2

Example 2:-

Which of the following is an arithmetic sequence?

1 - $T_n = 3n - 4$

2 - $T_n = 2n^2 + 1$

The solution:-

1- $T_{n+1} = 3(n+1) - 4 = 3n - 1$

$$T_{n+1} - T_n = 3n - 1 - (3n - 4) = 3 \text{ constant}$$

$\therefore$ It is an arithmetic sequence.

2- $T_{n+1} = 2(n+1)^2 + 1 = 2n^2 + 4n + 2 + 1 = 2n^2 + 4n + 3$

$$\begin{aligned} T_{n+1} - T_n &= 2n^2 + 4n + 3 - 2n^2 - 1 \\ &= 4n + 2 \neq \text{constant} \end{aligned}$$

$\therefore$ It is not an arithmetic sequence.

Note:-

The arithmetic sequence is a function of the first degree in n , and the coefficient of n is the common difference of the sequence.

In the previous example:-

- $T_n = 3n - 4$ is an arithmetic sequence its common difference = 3
- $T_n = 2n^2 + 1$ is not an arithmetic sequence because it is from the 2nd degree.

The Increasing sequence:-

It is a sequence its common difference is bigger than 0. such as (5 , 9 , 13 , 17 ,)

The Decreasing sequence:-

It is a sequence its common difference is smaller than 0. such as (11 , 9 , 7 , 5 ,)

Example 3:-

If : $2a + 1$, $5a - 1$, $6a + 3$ are three consecutive terms of an arithmetic sequence, find the value of a

The solution:-

$$(5a - 1) - (2a + 1) = (6a + 3) - (5a - 1)$$

$$3a - 2 = a + 4$$

$$2a = 6 \quad \therefore a = 3$$

Exercises (1)

1- If $(29, x, \dots, 3x, 95)$ is an arithmetic sequence, find

The value of x .

2- If $T_n = 15 - 3n$ is an arithmetic sequence, find its common difference.

3- In the arithmetic sequence $T_n = 5n + 2$, find the common difference and write the first three terms of this sequence.

4- Determine which of the following sequences are arithmetic:-

1) $(9, 13, 17, 21, \dots)$

2) $(4, 7, 12, 19, \dots)$

3) $(9, 9, 9, 9, \dots)$

4) $(-5, -11, -17, -23)$

Lesson two
The n^{th} term of the Arithmetic sequence

The n^{th} term of the Arithmetic sequence:-

$$T_n = a + (n - 1) d$$

Where:

- T_n : the value of the term
- n : the order of the term
- a : the first term of the sequence
- d : the common difference

The General form of the Arithmetic sequence:

Put $n = 1, 2, 3, 4, \dots$

$a, a + d, a + 2d, a + 3d, \dots, l$

Example (1)

In the arithmetic sequence (12, 14, 16, 18) find the value of the tenth term and the order of the term whose value equals 102.

The solution:-

$$a = 12, d = 2$$

$$T_{10} = a + 9d = 12 + 9 \times 2 = 30$$

$$a + (n-1)d = 102$$

$$12 + (n-1) \times 2 = 102$$

$$12 + 2n - 2 = 102$$

$$2n = 92$$

$$n = 46$$

Example (2)

Find number of terms of the arithmetic sequence (5, 9, 13,, 205)

The solution:-

$$a = 5 \qquad d = 4 \qquad l = 205$$

$$a + (n - 1) d = 205$$

$$5 + (n - 1) \times 4 = 205$$

$$5 + 4n - 4 = 205$$

$$4n = 204$$

$$n = 51 \qquad \therefore \text{the number of terms of the sequence} = 51$$

Example (3)

Find the order and the value of the first positive term of the sequence

(-42, -39, -36,, 21)

The solution:-

$$a = -42 \qquad , \quad d = 3 \qquad , \quad l = 21$$

$$a + (n - 1) d > 0$$

$$-42 + (n - 1) \times 3 > 0$$

$$-42 + 3n - 3 > 0$$

$$3n > 45$$

$$n > 15 \qquad n = 16$$

$$T_{16} = a + 15 d$$

$$T_{16} = -42 + 15 \times 3$$

$$T_{16} = 3$$

Exercises (2)

- 1- In the arithmetic sequence (63 , 59 , 55 , , -133),find the value of the eighth term and the number of terms of this sequence.
- 2- Find the order and the value of the first negative term of the arithmetic sequence
(11, 9 , 7 ,.....) (Put : $a + (n-1)d < 0$)
- 3- Find the order and the value of the first positive term of the arithmetic sequence
(- 51, -48 , -45 ,.....)
- 4- Prove that there is no term its value equals 100 in the arithmetic sequence
(13 , 17 , 21 ,).
- 5- In the arithmetic sequence (10 , 21 , 32 ,) ,find the order of the first term whose value bigger than 211.
- 6- Find the order of the term whose value is 99 in the arithmetic sequence
(3 , 5 , 7 ,)

Lesson (3)

Identifying the Arithmetic Sequence

Example (1)

T_n is an arithmetic sequence , $T_5 = 2$ and $T_7 = 8$, Find the sequence.

The solution:-

$$T_5 = 2$$

$$a + 4d = 2 \dots\dots\dots(1)$$

$$T_7 = 8$$

$$a + 6d = 8 \dots\dots\dots(2)$$

$$a + 6d = 8$$

$$-a + 4d = -2$$

$$2d = 6$$

$$d = 3$$

From (1) $a + 12 = 2$

$$a = -10$$

The sequence is (- 10 , - 7, - 4,)

Example (2)

If the sum of the second and the twelfth terms of an arithmetic sequence is 64 and the difference between its seventh term and the fourth term is 12. Find the sequence.

The solution:-

$$T_2 + T_{12} = 64$$

$$a + d + a + 11d = 64$$

$$2a + 12d = 64 \dots\dots\dots(1)$$

$$T_7 - T_4 = 12$$

$$a + 6d - a - 3d = 12$$

$$3d = 12$$

$$d = 4$$

From (1) $a + 6d = 32$

$$a + 24 = 32$$

$$a = 8$$

The sequence is (8 , 12 , 16 , 20 ,)

Example (3)

An arithmetic sequence its fourth term = 11 ,and the sum of its fifth and ninth terms = 40 find the sequence.

$$T_4 = 11$$

$$a + 3d = 11 \dots\dots\dots(1)$$

$$T_5 + T_9 = 40$$

$$a + 4d + a + 8d = 40$$

$$2a + 12d = 40 \dots\dots\dots(2)$$

By solving (1) and (2)

$$a = 2 \quad , \quad d = 3$$

The sequence is (2 , 5 , 8 , 11 ,)

Exercises (3)

- 1- T_n is an arithmetic sequence , $T_7 = 18$ and $T_{15} = 34$, Find the sequence.
- 2- Find the arithmetic sequence whose the sum of its second and fourth terms is 4 and the sum of its sixth, seventh and eighth terms is 54.
- 3- Find the arithmetic sequence whose the sum of its third and fourth terms is 7, and its ninth term is greater than its fifth term by 4.
- 4- An arithmetic sequence its first term = 5 , $T_3 = T_2 + 3$, find its fifth term.
- 5- In an arithmetic sequence If : $T_{17} = 73$, $T_{73} = 17$, find the sequence and the order of the term whose value = 0

Lesson (4) Arithmetic Means

- The arithmetic mean between the two terms a and b is $\frac{a+b}{2}$
- If a , b and c are three consecutive terms in an arithmetic sequence, then b is called the arithmetic mean between the two terms a and c
i.e.: $2b = a + c$

The General form of the Arithmetic sequence is

(a , $a + d$, $a + 2d$, $a + 3d$, , $l - 2d$, $l - d$, l)

↓
Arithmetic means

Note that:-

- The first mean is the second term.
- The last mean is term before the last.
- If the number of terms of a sequence = $n \rightarrow$ number of means = n

Example (1)

Insert 11 arithmetic means between 10 and 34.

The solution:-

$$a = 10 \quad , n = 13 \quad , T_{13} = 34$$

$$a + 12d = 34$$

$$10 + 12d = 34$$

$$12d = 24 \quad d = 2$$

The means are :

12 , 14 , 16 , 18 , 20 , 22 , 24 , 26 , 28 , 30 , 32

Example (2)

Two numbers the ratio between them is 1 : 3 and their arithmetic mean = 22, find the two numbers.

The solution:-

Let the two numbers are : $x = m$, $y = 3m$

$$X + y = 44$$

$$m + 3m = 44 \longrightarrow m = 11$$

The two numbers are 11 , 33

Example (3)

If (a , 5 , 11 , b , c) are consecutive terms of an arithmetic sequence, find the value of a , b and c.

The solution:-

$$a + 11 = 10 \longrightarrow a = -1$$

$$d = 11 - 5 = 6 \longrightarrow b = 17 , \quad c = 23$$

Example (4)

An arithmetic sequence its first term = 5 and its last term = 41, The ratio between the sum of the first two means and the sum of the last two means is 4 : 19 . find the sequence.

The solution:-

5 , 5 + d , 5 + 2d , , 41 - 2d , 41 - d , 41

$$\frac{5+d+5+2d}{41-d+41-2d} = \frac{4}{19} \longrightarrow \frac{10+3d}{82-3d} = \frac{4}{19}$$

$$190 + 57d = 328 - 12d \longrightarrow d = 2$$

The sequence is (5 , 7 , 9 ,)

Exercises (4)

- 1 - Insert 5 arithmetic means between 6 and 48.
- 2 - If 36 , a , 24 , b are consecutive terms of an arithmetic sequence, find the value of a and b.
- 3 - An arithmetic sequence its ninth term = 25 and the arithmetic mean between the third and fifth terms is 10 , find the sequence.
- 4- If the arithmetic mean of the two numbers x and 26 is 21 ,find the value of x.
- 5- If we insert number of arithmetic means between 2 , 47 ,and if the ratio between the second mean and the last mean is 2 : 7 . find these means.

Lesson (5)

The sum of the arithmetic sequence

The Sum of the First n Terms of an Arithmetic Sequence:-

$$S_n = \frac{n}{2} (a + l)$$

or

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

Example (1):-

(T_n) is an Arithmetic Sequence in which T₁ + T₃ = 10 , T₁₀ = 21 , find the sequence , then find the sum of the first twenty terms .

The solution:-

$$T_1 + T_3 = 10$$

$$T_{10} = 21$$

$$a + a + 2d = 10$$

$$a + 9d = 21 \dots\dots\dots(2)$$

$$2a + 2d = 10 \dots\dots\dots(1)$$

By solving (1) and (2) a = 3 , d = 2

The sequence is (3 , 5 , 7 ,)

$$S_{20} = \frac{n}{2} (2a + (n-1)d)$$

$$= 10 (6 + 19 \times 2) = 440$$

Example (2):

Find the number of the terms necessary to be taken from the sequence (9 , 12 , 15 ,) starting from the first term to get a sum equal to 306.

The solution:-

$$a = 9, d = 3$$

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$\frac{n}{2} (18 + 3n - 3) = 306$$

$$n(15 + 3n) = 612$$

$$3n^2 + 15n - 612 = 0$$

By factorization : $n = 12, n = -17$ refused

number of the terms necessary to be taken from the sequence to get a sum equal to 306
=12 terms

Example (3):

Find the sum of the first fifteen terms of the sequence (T_n) where $T_n = 5n + 1$

The solution:-

$$T_1 = 6 \longrightarrow a = 6, d = 5, n = 15$$

$$S_n = \frac{n}{2} (2a + (n-1)d) = \frac{15}{2} (12 + 14 \times 5) = 615$$

Example (4):

An arithmetic sequence its first term = 17, last term = -21, and the sum of its terms = -40.
find the sequence, then find the sum of the first ten terms.

The solution:-

$$S_n = \frac{n}{2} (a + l)$$

$$-40 = \frac{n}{2} (17 - 21)$$

$$-40 = -2n$$

$$n = 20$$

$$T_{20} = -21$$

$$a + 19d = -21$$

$$17b + 19d = -21$$

$$d = -2$$

the sequence is (17 , 15 , 13 ,)

$$S_n = \frac{n}{2} (2a + (n-1) d) = \frac{10}{2} (34 + 9 \times -2) = 80$$

Exercises (5)

1-In the arithmetic sequence (5 , 8 , 11 ,), Find the sum of the first 20 terms .

2- Find the sum of the first thirty terms of the sequence (T_n) where $T_n = 2n + 3$

3 - Find the number of the terms necessary to be taken from the sequence (1, 3 , 5 ,.....) starting from the first term to get a sum equal to 400.

4- Find the number of the terms necessary to be taken from the sequence (27 , 24, 21 , ...) starting from the first term to vanish the sum.

5- An arithmetic sequence Its first term = 12 and its last term = - 26 , the sum of the its terms = -140 . find the sequence.

Lesson (6)

Geometric sequence

The sequence (T_n) is called a geometric sequence if $\frac{T_{n+1}}{T_n} = a$ constant for each $n \in \mathbb{Z}^+$

This constant is called the common ratio of the sequence and is denoted by the symbol (r)

Example (1):-

Prove that the sequence (T_n) is a geometric sequence where $T_n = 10 \times 2^{n-1}$

The solution :-

$$T_{n+1} = 10 \times 2^n$$

$$\frac{T_{n+1}}{T_n} = \frac{10 \times 2^n}{10 \times 2^{n-1}} = 2 \text{ constant}$$

$\therefore$ The sequence is geometric and its common ratio $(r) = 2$

Example (2):-

Is the sequence (T_n) where $T_n = 5 + 3^n$ a geometric sequence?

The solution :-

$$T_{n+1} = 5 + 3^{n+1}$$

$$\frac{T_{n+1}}{T_n} = \frac{5 + 3^{n+1}}{5 + 3^n} \neq \text{constant}$$

$\therefore$ The sequence is not geometric

The geometric sequence whose first term is (a) and its common ratio (r).

❖ Is increasing if $r > 1, a > 0$ or $0 < r < 1, a < 0$

The geometric sequence (3 , 6 , 12 , 24 ,) Inc.

The geometric sequence (- 20 , - 10 , -5 ,) Inc.

any term is bigger than
the previous term

❖ Is decreasing if $r > 1, a < 0$ or $0 < r < 1, a > 0$

The geometric sequence (2 , 1 , $\frac{1}{2}$, $\frac{1}{4}$,) Dec.

The geometric sequence (- 2 , - 6 , -18 ,) Dec.

any term is smaller than
the previous term

❖ Is constant if $r = 1$

The geometric sequence (9 , 9 , 9 ,) is constant

Exercises (6)

1- Which of the following sequences is geometric :

- (2 , 5 , 8 , 11 ,)
- (3 . 33 . 3 . 33 ,)
- (4 , -4 , 4 , -4 ,)
- (1, 4 , 8 , 16 ,)

2 - Which of the following is geometric sequences:

- $T_n = 4n^3$ for every $n \geq 1$
- $T_n = 2^n - 1$ for every $n \geq 1$

3 – Complete: The next term of the geometric sequences (8 , 6 , $\frac{9}{2}$, $\frac{27}{8}$,) is

4- Choose the correct answer:-

The geometric sequences whose first term (a) and common ratio (r) is decreasing if.....

- 1) $a > 0$, $-1 < r < 0$
- 2) $a > 0$, $0 < r < 1$
- 3) $a < 0$, $-1 < r < 0$
- 4) $a < 0$, $0 < r < 1$

Lesson (7)

Identifying the Geometric Sequence

The n^{th} (General) term of the geometric sequence:-

$$T_n = a (r)^{n-1}$$

Example(1):-

Geometric sequence its third term = 9 and its sixth term = 243 , find the sequence.

The solution:-

$$T_3 = 9$$

$$a r^2 = 9 \dots\dots\dots(1)$$

By dividing (2) by (1)

$$r^3 = 27$$

from (1)

$$9a = 9$$

$$T_6 = 243$$

$$a r^5 = 243 \dots\dots\dots(2)$$

$$r = 3$$

$$a = 1$$

The sequence is (1 , 3 , 9 , 27 ,.....)

Example(2):-

Geometric sequence, the sum of the first and the second terms = 288 and the sum of the six and the seventh terms = 9, find the sequence.

The solution:-

$$T_1 + T_2 = 288$$

$$a + a r = 288$$

$$a (1 + r) = 288 \dots\dots\dots(1)$$

By dividing (2) by (1)

$$r^5 = \frac{1}{32}$$

$$\text{From (1) : } a \left(1 + \frac{1}{32} \right) = 288 \dots\dots\dots a = 192$$

$$T_6 + T_7 = 9$$

$$a r^5 + a r^6 = 9$$

$$a r^5 (1 + r) = 9 \dots\dots\dots(2)$$

$$r = \frac{1}{32}$$

The sequence is (192 , 96 , 48,.....)

Example(3):-

If (7 , x , y , 189) Geometric sequence . find the value of x , y

The solution:-

$$T_4 = 189$$

$$a r^3 = 189$$

$$7 r^3 = 189$$

$$r^3 = 27$$

$$r = 3$$

$$x = 21 \quad , \quad y = 63$$

Example(4):-

Find the order of the term whose value = 1024 in the geometric sequence

$$\left(\frac{1}{8} , \frac{1}{4} , \frac{1}{2} , 1, \dots \right)$$

The solution:-

$$a = \frac{1}{8} \quad , \quad r = 2$$

$$a r^{n-1} = 1024$$

$$\frac{1}{8} \times 2^{n-1} = 1024$$

$$2^{n-1} = 8192$$

$$2^{n-1} = 2^{13}$$

$$n - 1 = 13 \quad \dots \quad n = 14$$

The term whose value 1024 is T_{14}

Exercises (7)

- 1- Find the geometric sequence whose first its term = 9, and its sixth term = 288.
- 2- Find the geometric sequence whose common ratio = $\frac{1}{2}$ and its third term = 24 .
- 3- Find the geometric sequence in which $T_3 = 12$, $T_8 = 384$.
- 4- Find the geometric sequence whose its third term = 18 and its sixth term = 486.
- 5- Find the geometric sequence in which $T_2 + T_5 = 36$, $T_3 + T_6 = 18$.
- 6- If (54 , x , y , 2) Geometric sequence , find the value of x , y.

Lesson (8)

Geometric Means

- If a, b, c three numbers forms geometric sequence then:-

$$b^2 = a \times c$$

$$, b = \pm \sqrt{a c}$$

- The general form of the geometric sequence :

$$a, a r, a r^2, a r^3, \dots, \frac{l}{r^2}, \frac{l}{r}, l$$

↓
Geometric Means

- If the number of terms of a sequence = n , then number of means = $n-2$

Example (1):-

Insert three geometric means between 2, 32

The solution:-

$$a = 2, \quad T_5 = 32$$

$$T_5 = a r^4 = 32$$

$$2 \times r^4 = 32 \quad \dots \quad r^4 = 16 \quad \dots \quad r = \pm 2$$

The means are 4, 8, 16 or -4, 8, -16

Example (2):-

If $(x-1, x+2, 3x)$ Geometric sequence, find the value of x .

The solution:-

$$(x+2)^2 = 3x(x-1)$$

$$x^2 + 4x + 4 = 3x^2 - 3x$$

$$2X^2 - 7x - 4 = 0$$

$$(x - 4)(2x + 1) = 0 \quad \dots\dots\dots X = 4 \quad \text{or} \quad x = \frac{-1}{2}$$

Example (3):-

If several geometric means are inserted between 2 and 486 and if the sum of the last two means equals nine times the sum of the first two means. find these means.

$$\frac{486}{r^2} + \frac{486}{r} = 9(2r + 2r^2) \dots\dots\dots (\times r^2)$$

$$486 + 486r = 9r^2(2r + 2r^2)$$

$$486(1 + r) = 18r^3(1 + r) \dots\dots\dots (\div 18(1 + r))$$

$$r^3 = 27$$

$$r = 3$$

$$ar^{n-1} = 486$$

$$3^{n-1} = 243 = 3^5$$

$$n - 1 = 5$$

$$n = 6$$

$\therefore$ number of means = 4 means

Exercises (8)

1- Insert six geometric means between $\frac{1}{4}$ and 32.

2- Complete: The geometric mean of the two numbers : 4 , 16 is

3 – Complete: If the geometric mean of the two numbers 9 , y is 15 ,then y =

4- Insert five geometric means between 4 , 2916.

5- If several geometric means are inserted between 3 and 384, and if the product of the second and the last means = 2304, find the number of these means.

Lesson (8)
The sum of the Geometric sequence

Sum of (n) Terms of a Geometric Sequence

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

or

$$S_n = \frac{lr - a}{r - 1}$$

Example (1):-

Find the sum of the first eight terms of the geometric sequence (1 , 2 , 4 ,.....)

The solution:

$$a = 1, \quad r = 2, \quad n = 8$$

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{2^8 - 1}{2 - 1} = 255$$

Example (2):-

Find the sum of terms of the geometric sequence whose first term = 4 and its last term = 972 , if its common ratio =3

The solution:

$$S_n = \frac{lr - a}{r - 1} = \frac{972 \times 3 - 4}{3 - 1} = 1456$$

Example (3):-

Find the sum of the first seven terms of the geometric sequence whose its n^{th} term is

$$T_n = (-2)^{n-1}.$$

The solution:

$$T_1 = 1, \quad T_2 = -2$$

∴ This is a geometric sequence its first term = 1 and the common ratio = - 2

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{(-2)^7 - 1}{-2 - 1} = 43$$

Example (4):-

How many terms are to be taken from the geometric sequence (3 , 6 , 12 , ...) starting from its first term to make the sum of these terms = 189 ?

The solution:

$$a = 3, \quad r = 2$$

$$S_n = \frac{a(r^n - 1)}{r - 1} = 189$$

$$\frac{3(2^n - 1)}{2 - 1} = 189$$

$$3(2^n - 1) = 189$$

$$2^n - 1 = 63$$

$$2^n = 64 = 2^6$$

$$n = 6$$

number of terms should be taken from the sequence to make the sum 189 = 6 terms.

Exercises (9)

1- Find the sum of the first eight terms of the sequence(T_n) Where $T_n = 2(3)^{n-1}$

2 - Find the sum of the first five terms of the geometric sequence (1 , 3 , 9 ,.....)

3 – How many terms should be taken from the terms of the sequence (3 , 6, 12 ,.....)

To make the sum of these terms = 384.

4- Geometric sequence , its fourth term = 8 and its seventh term =64 ,find this sequences, then find the sum of the first 10 terms starting from the first term.

5- Prove that the sequence $T_n = 10 \times 2^{n-1}$ is geometric ,then find number of terms should be taken from the terms of the sequence starting from the first term to get the sum 2555.

6 – Find the sum of the first ten terms of the Geometric sequence whose first term =2

And its common ratio = 1

7) Find the sum of the first ten terms of the Geometric sequence If $T_1 = 4 T_3$ and

$$T_2 + T_5 = 3$$

Lesson (10)
Sum of infinite Geometric sequence

We can sum an infinite number of terms of geometric sequence when $-1 < r < 1$

Then ,
$$S_{\infty} = \frac{a}{1-r}$$

Example (1):-

(T_n) is a geometric sequence in which $T_2 = 240$ and $T_5 = 30$. Find the sequence , Then find The sum of an infinite number of its terms.

The solution:-

$a r = 240$ (1)

$a r^4 = 30$ (2)

By dividing (2) $\div$ (1)

$$r^3 = \frac{1}{8} \longrightarrow r = \frac{1}{2}$$

from (1) $a = 480$

$$S_{\infty} = \frac{a}{1-r} = \frac{480}{1-0.5} = 960$$

Example (2):-

Geometric sequence in which $T_2 = 5$ and $\frac{T_7}{T_4} = \frac{1}{27}$ find the sequence and show that there are an infinite number of its terms can be summed and find this sum .

The solution:-

$$ar = 5 \dots\dots\dots(1)$$

$$\frac{ar^6}{ar^3} = \frac{1}{27}$$

$$r^3 = \frac{1}{27} \dots\dots\dots r = \frac{1}{3}$$

$$\text{From (1) } \dots\dots\dots a = 15$$

The sequence is (15 , 5 , $\frac{5}{3}$,)

$\because -1 < r < 1$ We can sum an infinite number of terms of this sequence

$$S_{\infty} = \frac{a}{1-r} = \frac{15}{1-\frac{1}{3}} = 22.5$$

Example (3):-

Find the geometric sequence whose terms are positive, The sum of its first and second terms = 16 , and the sum of an infinite number of its terms = 25.

The solution:-

$$T_1 + T_2 = 16$$

$$a + ar = 16$$

$$a(1 + r) = 16 \dots\dots\dots(1)$$

$$\frac{a}{1-r} = 25$$

$$a = 25(1 - r) \dots\dots\dots(2)$$

By substitution from (2) .. in (1)

$$25(1 - r)(1 + r) = 16$$

$$1 - r^2 = \frac{16}{25}$$

$$r^2 = \frac{9}{25}$$

$$r = \pm \frac{3}{5}$$

from (1) $a \left(1 + \frac{3}{5} \right) = 16$

$$a = 10$$

The sequence is $\left(10, 6, \frac{18}{5}, \dots \right)$

Exercises (10)

1- If the first term of a geometric sequence of an infinite number of terms = 18 and the fourth term = $\frac{16}{3}$. What is the sum of this sequence?

2- If the sum of an infinite geometric sequence is 54, and its first term = 18. Find the sequence.

3- If the first term of a geometric sequence = 81, and its sixth term = $\frac{1}{3}$. Find the sum of an infinite number of its terms starting from its first term.

4- Find the sum of an infinite number of terms of the sequence $(8, 4, 2, \dots)$.

5- If the sum of an infinite geometric sequence whose first term = 12 is 96, find its common ratio.

6- Find the sum of an infinite number of terms of the geometric sequence : $T_n = 3^{3-n}$.

Unit Test

Question (1)

- 1- If $(29, x, \dots, 3x, 95)$ an arithmetic sequence .Find the value of x .
- 2- The sum of an infinite geometric sequence is 54 ,and its first term = 18 .find the sequence.

Question (2)

- 1- Prove that there is no term its value equals 100 in the arithmetic sequence $(13, 17, 21, \dots)$.
- 2- Find the sum of the first ten terms of the geometric sequence whose first term $a = 2$ and its common ratio $r = 1$.

Question (3)

- 1- In an arithmetic sequence If : $T_{17} = 73$, $T_{73} = 17$, find the sequence and the order of the term whose value equals zero.
- 2- Insert five geometric means between 4 , 2916.

Question (4)

- 1 – If the arithmetic mean between the two numbers x , 26 is 21 .find the value of x .
- 2- Find the geometric sequence whose first term = 9 and its sixth term = 288.

Question (5)

- 1-Find the number of the terms necessary to be taken from the sequence $(27, 24, 21, \dots)$ starting from the first term to vanish the sum.
- 2- Find the geometric sequence in which $T_3 = 12$, $T_8 = 384$.

Evaluation

Applied Technology School

Unit (2)
Complex Numbers

By the end of this Unit student should be able to:

- ❖ Determine the modulus and the amplitude of the complex number.
- ❖ Identify the principle amplitude of a complex number.
- ❖ Identify the trigonometric form of a complex number.
- ❖ Identify the modulus and the amplitude of the product and the quotient of two complex numbers.
- ❖ Identify the Exponential form of a complex number (Euler's rule).
- ❖ Identify the multiplication and division of complex numbers by using exponential form.
- ❖ Identify De Moivre's theorem and its applications.
- ❖ Deduce the n^{th} roots of any complex number.
- ❖ Identify the cubic roots of unity.
- ❖ Solve Application problems on the cubic of unity.

Lesson 1

The modulus and the amplitude of the complex number

If $z = x + yi$ is a complex number represented by point (x, y) in Argand's plane. Then the modulus is the distance from the origin and denoted by $|z|$ or r and θ is called amplitude of the complex number

$$r = \sqrt{x^2 + y^2}, \quad \tan \theta = \frac{y}{x} \quad \text{thus } \theta = \tan^{-1} \frac{y}{x} \quad \theta \in \left] \frac{-\pi}{2}, \frac{\pi}{2} \right[$$

Example 1:

Find the modulus and the amplitude of the complex number $z = 1 - \sqrt{3}i$

The Solution:

$x = 1, y = -\sqrt{3}$ in the fourth quadrant

$$\text{Then } r = \sqrt{1 + 3} = 2, \quad \theta = \tan^{-1} \frac{-\sqrt{3}}{1} = -60^\circ = \frac{-\pi}{3}$$

Example 2:

Find the modulus and the amplitude of the complex number $z = 1 + i$

The Solution:

$x = 1, y = 1$ in the first quadrant

$$\text{Then } r = \sqrt{1 + 1} = \sqrt{2}, \quad \theta = \tan^{-1} \frac{1}{1} = 45^\circ = \frac{\pi}{4}$$

Example 3:

Find the modulus and the amplitude of the complex number $z = -\sqrt{3} + i$

The Solution:

$x = -\sqrt{3}$, $y = 1$ in the second quadrant

Then $r = \sqrt{3 + 1} = 2$, $\theta = \tan^{-1} \frac{1}{-\sqrt{3}} = 150^\circ$

Use the calculator to find modulus and the amplitude of complex number:

Example4 :

$$z = -1 - i$$

Mode 2	-1	-1	ENG	=	shift	2	3	=
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Result:

$$r = \sqrt{2} , \theta = -135 = -\frac{3\pi}{4}$$

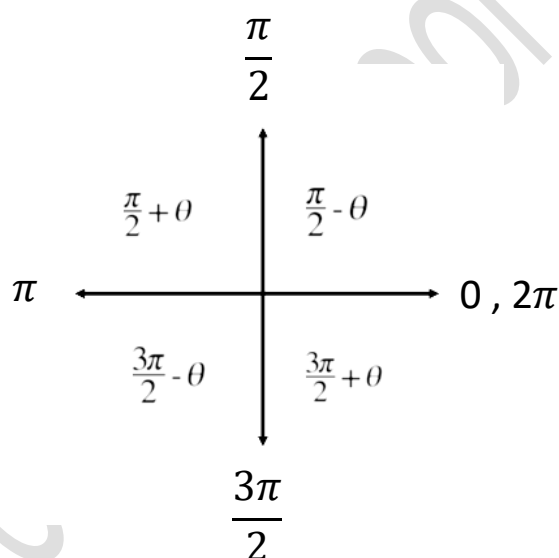
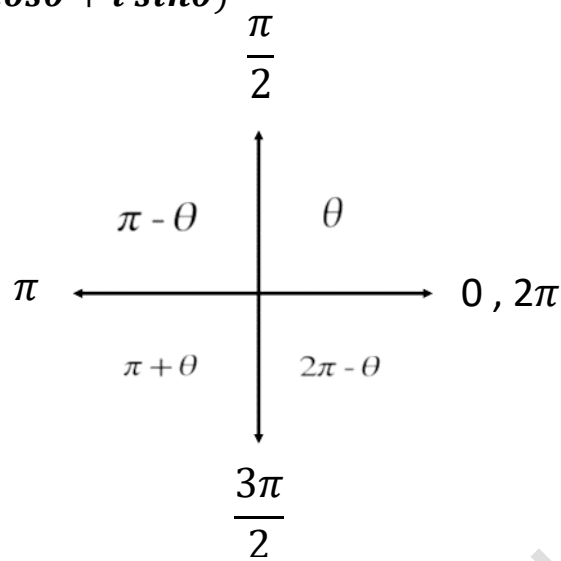
Exercises (1)

- 1- If $z = 3 + 4i$, then find $|z|$
- 2- Find amplitude of complex number $z = -2i$
- 3 - Find the modulus and amplitude for complex number $z = -3$
- 4- Find the amplitude for complex number $z = 1 - i$
- 5- If $z_1 = 2i$, $z_2 = -1 + 3i$ such that $i^2 = -1$, find amplitude of $(z_1 - z_2)$
- 6- If $z_1 = 3 + 3\sqrt{3}i$, $z_2 = -4 - 4\sqrt{3}i$, find amplitude of $(z_1 + z_2)$

Lesson(2)

The trigonometric form of complex number

The trigonometric form of complex number that modulus is r and amplitude is θ is $z = r(\cos\theta + i \sin\theta)$



Example 1:

Write in trigonometric form for complex number $z = \frac{-4}{\sqrt{3}+i}$

Solution :

$$z = \frac{-4}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} = -\sqrt{3} + i$$

$$r = 2, \theta = \frac{5\pi}{6}$$

$$z = 2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$$

Example 2:

Write in trigonometric form for complex number $z = -3 - 3i$

Solution:

$$r = 3\sqrt{2} \quad , \quad \theta = -135 = -\frac{3\pi}{4}$$

$$z = 3\sqrt{2} \left(\cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right) \right)$$

Example 3:

Write in trigonometric form for complex number $z = -2i$

Solution:

$$r = 2 \quad , \quad \theta = -90 = \frac{\pi}{2}$$

$$z = 2 \left(\cos\left(\frac{-\pi}{2}\right) + i \sin\left(\frac{-\pi}{2}\right) \right)$$

Example 4:

If $z = \sqrt{2}(\sin 30 + i \cos 30)$, then find the amplitude for z

Solution:

The function is incorrect (you can convert z to algebraic form then find the amplitude)

$$z = \frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2}i$$

$$\theta = 60 = \frac{\pi}{3}$$

Example 5:

Find the modulus and the amplitude for the complex number

$$z = -8 (\cos 45 + i \sin 45)$$

Solution:

The modulus is incorrect (we can you can convert z to algebraic form)

$$z = -4\sqrt{2} - 4\sqrt{2}i$$

$$r = 8, \theta = -135 = \frac{-3\pi}{4}$$

Exercises (2)

[1] Find the trigonometric form for each of the following:

- $z = -\sqrt{3}i$
- $z = 1 + i$
- 3) $z = -1 + i$
- 4) $z = -1 + \sqrt{3}i$
- 5) $z = 2 + 2\sqrt{3}i$
- 6) $z = -3 - 3\sqrt{3}i$
- 7) $z = \frac{5-3\sqrt{3}i}{1+2\sqrt{3}i}$ such that $i^2 = -1$

[2] Find the amplitude for each of the following:

- ❖ $z = \cos \frac{7\pi}{4} - i \sin \frac{7\pi}{4}$
- ❖ $z = \cos \frac{5\pi}{3} - i \sin \frac{5\pi}{3}$
- ❖ $z = \sin \frac{\pi}{6} - i \cos \frac{\pi}{6}$
- ❖ $z = \sin 30 + i \cos 30$

Lesson 3
Multiplying complex number in trigonometric form

If $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$$

If $z = r(\cos \theta + i \sin \theta)$, then $z^n = r^n(\cos n\theta + i \sin n\theta)$

Example 1:

If $z_1 = 3(\cos 75 + i \sin 75)$, $z_2 = 4(\cos 15 + i \sin 15)$, then find $z_1 z_2$ in the form of $x + y i$

Solution:

$$z_1 z_2 = 12(\cos 90 + i \sin 90) = 12 i$$

Example 2:

$z_1 = 2(\cos 100 + i \sin 100)$, $z_2 = 4(\cos 340 + i \sin 340)$, then find $z_1 z_2$

Solution:

$$z_1 z_2 = 8(\cos 440 + i \sin 440) = 8(\cos 80 + i \sin 80)$$

Example 3:

Put the number $z = (1 - i)$ in the trigonometric form , then find $(1 - i)^8$

Solution:

$$z = \sqrt{2}(\cos -45 + i \sin -45)$$

$$z^8 = 16(\cos -360 + i \sin -360) = 16$$

Example 4:

If $z_1 = \cos \frac{7\pi}{4} - i \sin \frac{7\pi}{4}$, $z_2 = -1 + i$, then find $z = z_1^2 z_2^6$ in the algebraic form

Solution:

$$z_1 = \cos(-45) + i \sin(-45)$$

$$z_1^2 = \cos(-90) + i \sin(-90)$$

$$z_2 = \sqrt{2} \cos(135) + i \sin(135)$$

$$z_2^6 = 8 \cos(90) + i \sin(90)$$

$$z_1^2 z_2^6 = 8(\cos 0 + i \sin 0) = 8$$

Exercises (3)

- 1- Find in the trigonometric form : $3(\cos 30 + i \sin 30) \times 6(\cos 70 + i \sin 70)$
- 2- Find in the trigonometric form : $(5 (\cos 10 + i \sin 10))^2$
- 3- If $x = \cos 17 + i \sin 17$, $y = \cos 11 + i \sin 11$, then find $z = x^3 y^9$ in the trigonometric form
- 4- If $z_1 = \cos 70 + i \sin 70$, $z_2 = \cos 20 + i \sin 20$, then find $z = z_1 z_2$ in the trigonometric form
- 5- If $z_1 = \cos 75 + i \sin 75$, $z_2 = \cos 15 + i \sin 15$, then find $z = z_1 z_2$ in the trigonometric form
- 6- If $z_1 = \cos 75 + i \sin 75$, $z_2 = \cos 15 + i \sin 15$, then find $z = z_1 + z_2$ in the trigonometric form

Lesson 4

Dividing complex number in trigonometric form

If $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2))$$

Example 1:

$$z_1 = 2(\cos 100 + i \sin 100) , z_2 = 4(\cos 340 + i \sin 340) ,$$

find $\frac{z_2}{z_1}$ in the form of $x + iy$

Solution:

$$\frac{z_2}{z_1} = 2(\cos 240 + i \sin 240) = -1 - i\sqrt{3}$$

Example 2:

Put the numbers $i\sqrt{2}$, $1 + i$ in the trigonometric form , then use it to find the value of $(\frac{i\sqrt{2}}{1+i})^{12}$

Solution:

$$i\sqrt{2} = \sqrt{2}(\cos 90 + i \sin 90)$$

$$1 + i = \sqrt{2}(\cos 45 + i \sin 45)$$

$$\frac{i\sqrt{2}}{1+i} = (\cos 45 + i \sin 45)$$

$$(\frac{i\sqrt{2}}{1+i})^{12} = (\cos 180 + i \sin 180) = -1$$

Example 3:

$$\text{If } z_1 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$$

$$z_2 = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$$

Find the amplitude of the number $z = \left(\frac{z_1}{z_2}\right)^7$

Solution:

$$\frac{z_1}{z_2} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

$$\left(\frac{z_1}{z_2}\right)^7 = \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}$$

The amplitude of the number $z = -\frac{\pi}{2}$

Exercises (4)

1 - Find $6(\cos 210 + i \sin 210) \div 3(\cos 70 + i \sin 70)$

2 - If $z_1 = 6(\cos 240 + i \sin 240)$

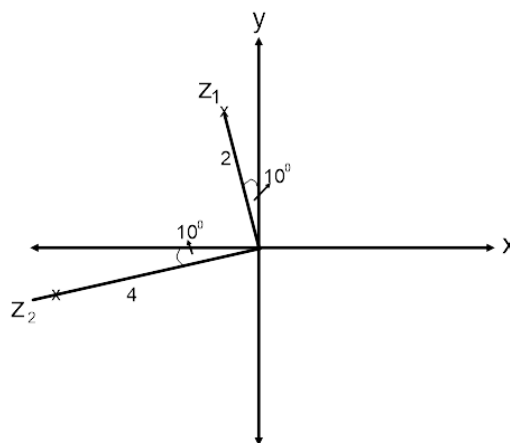
$z_2 = 4(\cos 300 + i \sin 300)$, find $\frac{z_2}{z_1}$ in the trigonometric form

3 - If $z_1 = 10(\cos \theta + i \sin \theta)$, $z_2 = (5 \cos \alpha + i \sin \alpha)$

And $\theta - \alpha = \frac{3\pi}{4}$, find $\frac{z_1}{z_2}$ in the form of $x + iy$

4 - In the opposite figure :

Find in trigonometric form $\frac{z_1}{z_2}$



Lesson 5

Exponential form of a complex number (Euler form)

The exponential form of a complex number (Euler form):

If $z = x + i y$ (Algebraic form)

$= r(\cos \theta + i \sin \theta)$ (Trigonometric form)

Then , the exponential form of z is $z = r e^{\theta i}$, where θ is the =radian measure

Example 1:

Find the exponential form of a complex number $z = \frac{-4}{\sqrt{3}+i}$

Solution:

$$z = \frac{-4}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} = -\sqrt{3} + i$$

$$r = 2 \quad , \quad \theta = \frac{5\pi}{3}, \quad z = 2 e^{\frac{5\pi}{3}i}$$

Example 2:

Find the exponential form of a complex number $z = -3 - 3i$

Solution:

$$r = 3\sqrt{2} \quad , \quad \theta = -135 = \frac{3\pi}{4}$$

$$z = 3\sqrt{2} e^{\frac{-3\pi}{4}i}$$

Example 3:

Find the exponential form of the complex number $z = -2i$

Solution:

$$r = 2, \theta = -90 = \frac{-\pi}{2}$$

$$z = 2e^{\frac{-\pi}{2}i}$$

Example 4 :

If $z = \sqrt{2}(\sin 30 + i \cos 30)$, then find the exponential form of z

Solution:

The function is incorrect (you can convert z to algebraic form then find the amplitude)

$$z = \frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2}i, \theta = 60 = \frac{\pi}{3}$$

$$z = 2e^{\frac{\pi}{3}i}$$

Example 5:

Find the modulus and the amplitude of the complex number

$z = -8(\cos 45 + i \sin 45)$ then write its exponential form

Solution:

The modulus is incorrect (you can convert z to algebraic form)

$$z = -4\sqrt{2} - 4\sqrt{2}i$$

$$r = 8, \theta = -135 = \frac{-3\pi}{4}$$

$$z = 8e^{\frac{-3\pi}{4}i}$$

Exercises (5)

[1] Find the exponential form of each of the following complex number:

- $z = 1 - \sqrt{3}i$
- $z = 1 + i$
- $z = 2\sqrt{2}(1 + i)$
- $z = -1 + i$
- $z = -1 + \sqrt{3}i$
- $z = 2 + 2\sqrt{3}i$
- $z = -3 - 3\sqrt{3}i$
- 8) $z = \frac{5-3\sqrt{3}i}{1+2\sqrt{3}i}$ where $i^2 = -1$

[2] Find the exponential form of each of the following complex number:

$$\diamond z = \cos \frac{7\pi}{4} - i \sin \frac{7\pi}{4}$$

$$\diamond z = \cos \frac{5\pi}{3} - i \sin \frac{5\pi}{3}$$

$$\diamond z = \sin \frac{\pi}{6} - i \cos \frac{\pi}{6}$$

$$\diamond z = \sin 30 + i \cos 30$$

Lesson (6)
Multiplying complex number using the exponential form

If $z_1 = r_1 e^{\theta_1 i}$ $z_2 = r_2 e^{\theta_2 i}$ then

$$z_1 z_2 = r_1 r_2 e^{(\theta_1 + \theta_2)i}$$

Example 1:

If $z_1 = 3(\cos 60 + i \sin 60)$, $z_2 = 4(\cos 30 + i \sin 30)$

Find $z_1 z_2$ in the exponential form

Solution:

$$z_1 z_2 = 12(\cos 90 + i \sin 90) = 12 e^{\frac{\pi}{2}i}$$

Example 2:

If $z_1 = 2(\cos 100 + i \sin 100)$, $z_2 = 4(\cos 340 + i \sin 340)$

Find $z_1 z_2$ in the exponential form

Solution:

$$z_1 z_2 = 8(\cos 440 + i \sin 440) = 8(\cos 80 + i \sin 80) = 8 e^{\frac{4\pi}{9}i}$$

Example (3) :

Put the number $z = 1 - i$ in the exponential form

Solution:

$$z = \sqrt{2}(\cos(-45) + i \sin(-45))$$

$$z = \sqrt{2}e^{\frac{-\pi}{4}i}$$

Example 4 :

$$\text{If } z_1 = \cos \frac{7\pi}{4} - i \sin \frac{7\pi}{4}, \quad z_2 = -1 + i$$

Find $z_1 z_2$ in the exponential form

Solution:

$$z_1 = \cos(-45) + i \sin(-45) = e^{\frac{-\pi}{4}i}$$

$$z_2 = \sqrt{2} \cos 135 + i \sin 135 = \sqrt{2} e^{\frac{3\pi}{4}i}$$

$$z_1 z_2 = \sqrt{2} e^{\frac{\pi}{2}i}$$

Exercises(6)

- 1 - Put in the exponential form $3(\cos 30 + i \sin 30) \times 6(\cos 70 + i \sin 70)$
- 2- If $x = \cos 17 + i \sin 17$, $y = \cos 13 + i \sin 13$, find $z = z_1 z_2$ in the exponential form
- 3- If $z_1 = \cos 40 + i \sin 40$, $z_2 = \cos 50 + i \sin 50$, find $z = z_1 z_2$ in the exponential form
- 4 - If $z_1 = \cos 25 + i \sin 25$, $z_2 = \cos 95 + i \sin 95$, find $z = z_1 z_2$ in the exponential form
- 5- If $z_1 = 10(\cos \theta + i \sin \theta)$, $z_2 = 5(\cos \alpha + i \sin \alpha)$, and $\theta - \alpha = \pi$, find $z_1 z_2$

Lesson 7

Dividing the complex number using the exponential form

If $z_1 = r_1 e^{\theta_1 i}$ $z_2 = r_2 e^{\theta_2 i}$ then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{(\theta_1 + \theta_2)i}$$

Example 1:

If $z = \frac{i}{\cos 75 + i \sin 75}$, find z in the exponential form

Solution:

$$z = \frac{e^{\frac{\pi}{2}i}}{e^{\frac{5\pi}{12}i}} = e^{\frac{\pi}{12}i}$$

Example 2:

If $z_1 = \cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}$ and $|z_1| = 4$, the amplitude of $z_1 z_2 = \frac{-2\pi}{3}$, find

$\frac{z_1}{z_2}$ in the exponential form

Solution:

The amplitude of $z_1 z_2 = \text{amp}(z_1) + \text{amp}(z_2)$

$$\frac{-2\pi}{3} = \frac{7\pi}{6} + \text{amp}(z_2)$$

$$\text{amp}(z_2) = \frac{-11\pi}{6}$$

$$\frac{z_1}{z_2} = \frac{1}{4} e^{\frac{5\pi}{6}i}$$

Example 3:

If $z_1 = \cos 120 + i \sin 120$, $z_2 = \cos 30 + i \sin 30$

Find $z = \frac{z_1}{z_2}$ in the exponential form

Solution:

$$\frac{z_1}{z_2} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \quad (\text{Trigonometric form})$$

$$= e^{\frac{\pi}{2}i} \quad (\text{Exponential form})$$

Exercises (7)

1) find $6(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}) \div 3(\cos \frac{7\pi}{18} + i \sin \frac{7\pi}{18})$ in the exponential form

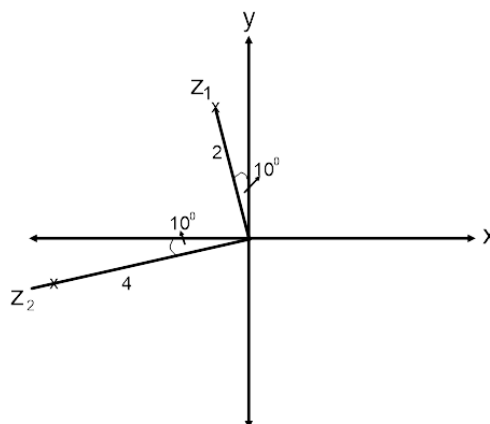
2) If $z_1 = 6 \cos 240 + i \sin 240$, $z_2 = 4(\cos 300 + i \sin 300)$

Find $\frac{z_2}{z_1}$ in the exponential form

3) If $z_1 = 10(\cos \theta + i \sin \theta)$, $z_2 = 5(\cos \alpha + i \sin \alpha)$, and $\theta - \alpha = \frac{3\pi}{4}$, find $\frac{z_1}{z_2}$

4) In the opposite figure :

Find in the exponential form $\frac{z_1}{z_2}$



Lesson 8

De Moivre's theorem with integer power

De Moivre's theorem (for positive integer power)

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

Example 1:

Find the trigonometric and the exponential form of $z = (i^{21})^3$ where $i^2 = -1$

Solution:

$$\begin{aligned} z &= i^{63} \\ &= (\cos 90 + i \sin 90)^{63} \\ &= \cos 5670 + i \sin 5670 = \cos 270 + i \sin 270 = e^{\frac{3\pi}{2}i} \end{aligned}$$

Example 2:

If $z = (1 + \cos \frac{\pi}{3} + i \sin \frac{\pi}{3})^6$, find z in the trigonometric and exponential form

Solution:

$$\begin{aligned} z &= \left(\frac{3}{2} + \frac{\sqrt{3}}{2}i\right)^6 \\ &= \left[\sqrt{3} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)\right]^6 \\ &= 27 (\cos \pi + i \sin \pi) \\ &= 27e^{\pi i} \end{aligned}$$

Exercises (8)

- 1 - Put $z = (1 - i)^8$ in the trigonometric form and the exponential form
- 2 - If $z_1 = \cos \frac{7\pi}{4} - i \sin \frac{7\pi}{4}$, $z_2 = -1 + i$, find $z_1^2 z_2^6$ in the algebraic form , trigonometric form and exponential form.
- 3 - Put $z = \left(\frac{5-3\sqrt{3}i}{1+2\sqrt{3}i} \right)^3$, in the trigonometric form and exponential form .
- 4 - If z is a complex number where $z = \left(\sin \frac{\pi}{6} - i \cos \frac{\pi}{6} \right)^4$, find the exponential form of z
- 5 - If $z = \sin \frac{\pi}{9} + i \cos \frac{\pi}{9}$, find z^9 in the exponential form
- 6 - Find the value of the number $(-1 + \sqrt{3}i)^{48}$ where $i^2 = -1$ in the algebraic form.

Lesson (9)

Demoivre's theorem with rational power

Demoivre's theorem with rational power:

$$(\cos \theta + i \sin \theta)^{\frac{1}{k}} = \cos \frac{\theta + 2r\pi}{k} + i \sin \frac{\theta + 2r\pi}{k}$$

Where: $r = \dots, -2, -1, 0, 1, 2, \dots$

Example (1):

Find in the trigonometric and exponential forms the roots of the following equation in the set of complex numbers: $Z^4 = 8(1 - \sqrt{3}i)$

The solution:-

$$Z^4 = 16 \left(\cos \frac{-\pi}{3} + i \sin \frac{-\pi}{3} \right)$$

$$\therefore Z = 2 \left(\cos \frac{\frac{-\pi}{3} + 2r\pi}{4} + i \sin \frac{\frac{-\pi}{3} + 2r\pi}{4} \right)$$

$$\text{When } r = 0, \text{ then } Z = \left(\cos \frac{-\pi}{12} + i \sin \frac{-\pi}{12} \right) = 2 e^{\frac{-\pi i}{12}}$$

$$\text{When } r = 1, \text{ then } Z = \left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right) = 2 e^{\frac{5\pi i}{12}}$$

$$\text{When } r = -1, \text{ then } Z = \left(\cos \frac{-7\pi}{12} + i \sin \frac{-7\pi}{12} \right) = 2 e^{\frac{-7\pi i}{12}}$$

$$\text{When } r = 2, \text{ then } Z = \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) = 2 e^{\frac{11\pi i}{12}}$$

Example (2):

Find The two square roots of the number (i) in the exponential form.

The solution:-

$$Z = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

$$= e^{\frac{\pi}{2}i}$$

$$Z^{\frac{1}{2}} = e^{\frac{\frac{\pi}{2} + 2\pi r}{2}i}$$

When $r = 0$,then : the first root = $e^{\frac{\pi}{4}i}$

When $r = 1$,then: the second root = $e^{\frac{-3\pi}{4}i}$

Exercises(9)

1- Find the two square roots of the number Z where $Z = -2i$, in the exponential form.

2- If $x = \sqrt{3} - i$, $y = \sqrt{3} + i$ and $Z = x^2 - 2xy + y^2$, Find the cubic roots of the number Z in the exponential form.

3- If $Z = \frac{2(5 - 3\sqrt{3}i)}{1 + 2\sqrt{3}i}$, where $i^2 = -1$, find $\sqrt{Z}$ in the exponential form.

4 - Find the two square roots of the number $Z = 2 - 2\sqrt{3}i$ in the two trigonometric and exponential forms.

5- If $Z_1 = \frac{6 + 4i}{1 + i}$, $Z_2 = \frac{26}{5 - i}$ and $Z = 4(Z_1 - Z_2)$. Find the cubic roots of the number Z .

Lesson (10)

The cubic roots of Unity

By using Demoivre's theorem we find that:-

The solution set of the equation :- $Z^3 = 1$

Is $\boxed{1}$, $\boxed{-\frac{-1}{2} - \frac{\sqrt{3}}{2} i}$, $\boxed{-\frac{-1}{2} + \frac{\sqrt{3}}{2} i}$

We can let that $1, \omega, \omega^2$ are the cubic roots of one.

The properties of the cubic roots of one:

1- $1 + \omega + \omega^2 = 0$

$$1 + \omega = -\omega^2$$

$$\omega + \omega^2 = -1$$

2- $\omega^3 = 1$

$$\omega^2 = \frac{1}{\omega}$$

$$\omega = \frac{1}{\omega^2}$$

3- $\omega - \omega^2 = \pm \sqrt{3} i$, $\omega^2 - \omega = \pm \sqrt{3} i$

Example(1):-

Find the value of $5 + 5\omega + 5\omega^2$

The solution:-

$$5 + 5\omega + 5\omega^2 = 5(1 + \omega + \omega^2) = 0$$

Example (2):

Find the value of:-

$$\left(1 - \frac{1}{\omega} - \frac{1}{\omega^2}\right)(3 + 5\omega + 5\omega^2)$$

The solution:-

$$\begin{aligned}(1 - \omega^2 - \omega)(3 + 5\omega + 5\omega^2) &= (1 + 1)(3 - 5) \\ &= 2 \times -2 = -4\end{aligned}$$

Example (3):

Find the value of: $(2 + 5\omega + 2\omega^2)^5$

The solution:-

$$\begin{aligned}(2 + 5\omega + 2\omega^2)^5 &= (5\omega - 2\omega)^5 \\ &= (3\omega)^5 = 243\omega^2\end{aligned}$$

Example (4):

Find the conjugate of the number (ω)

The solution:-

$$\text{The conjugate of the number } (\omega) = \omega^2$$

Example (5):

Find the value of $1 + \omega + \omega^2 + \omega^3 + \dots + \dots + \dots + \omega^{100}$

The solution:-

$$\text{The Expression} = 1 + \omega = -\omega^2$$

Exercises (10)

Complete the following:-

1 – The conjugate of the number $\omega + i$ is

2- $(1 + \omega + \omega^2)^3 = \dots\dots\dots$

3 – $(1 + \omega)^2 (\omega + \frac{1}{\omega}) = \dots\dots\dots$

4 – $(1 + 2\omega + \frac{1}{\omega^2})(1 + 2\omega + \frac{1}{\omega^4}) = \dots\dots\dots$

5 - $\omega + \omega^2 + \omega^3 + \dots\dots + \dots\dots + \omega^{100} = \dots\dots\dots$

6 – $(1 + \omega)^4 + (1 + \omega^2)^4 + (\omega + \omega^2)^4 = \dots\dots\dots$

Unit test

Question (1):-

1-Find the principal amplitude of the number $Z = 1 - i$.

2- Find $6 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right) \div 3 \left(\cos \frac{7\pi}{18} + i \sin \frac{7\pi}{18} \right)$ in the exponential form $\frac{Z_2}{Z_1}$.

Question (2):-

1- Write the trigonometric form of the complex number $Z = \frac{-4}{\sqrt{3} + i}$.

2- If $Z_1 = 3 (\cos 60^\circ + i \sin 60^\circ)$, $Z_2 = 4 (\cos 30^\circ + i \sin 30^\circ)$. Find $Z_1 Z_2$ in the exponential form.

Question (3):

1- If $Z_1 = (\cos 70^\circ + i \sin 70^\circ)$, $Z_2 = (\cos 20^\circ + i \sin 20^\circ)$ Find the number $Z = Z_1 Z_2$

In the trigonometric form.

2-Find the number $Z = \left(\frac{5-3\sqrt{3}i}{1+2\sqrt{3}i} \right)^3$ In the trigonometric form.

Question (4):

1- If $Z_1 = 10 (\cos \theta + i \sin \theta)$, $Z_2 = 5 (\cos \alpha + i \sin \alpha)$, If $\theta - \alpha = \frac{3\pi}{4}$, Find $\frac{Z_1}{Z_2}$ in the form $x + iy$.

2- Find the two square roots of the number Z where $Z = -2i$, in the exponential form.

Question 5:-

1- Find the Exponential form of the number Z where: $Z = \cos \frac{7\pi}{4} - i \sin \frac{7\pi}{4}$.

2- Find the value of $(2 + 5\omega + 2\omega^2)^5$

Evaluation

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